**Lab 18**

# Adding a Workflow Extension

*WSO2 API Manager 4.7.0 · Practical exercise*

## Before You Begin

URLs in this guide use placeholders. Replace them with the values of the API Manager you are using:

- **`<APIM_HOST>`** – host name or IP address of the API Manager server (localhost when API Manager runs on your own laptop).
- **`<HTTPS_PORT>`** – API Manager HTTPS port: 9443 plus any port offset (Management Console, Developer Portal, Admin Portal).

The server uses a self-signed certificate by default, so the browser shows a security warning the first time: choose Advanced and continue.

## Training Objective

Learn how to add a workflow extension for application creation, so that every new application must be approved before it can be used.

## Business Scenario

PizzaShack wants to control which applications are created in its Developer Portal: every new application must be approved by an administrator before it can be used.

## High-Level Steps

- Switch application creation to the approval workflow
- Create an application in the Developer Portal
- Approve (or reject) it in the Admin Portal and check the result
- Switch back to automatic approval

## Prerequisites

- API Manager is running and you can log in as admin.
- No restart is needed in this lab.

## Part 1 – Engage the Approval Workflow

**1.** Log in to the Management Console as admin and click **Main** → **Registry** → **Browse**.

**Management Console URL**

```
https://<APIM_HOST>:<HTTPS_PORT>/carbon
```

**2.** In the **Location** field, enter the path below, press Enter, and click **Edit as text**.

**Registry path**

```
/_system/governance/apimgt/applicationdata/workflow-extensions.xml
```

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3. Find the two `ApplicationCreation` lines near the top of the file. In the file each one is a single line; they are shown split here so they fit on the page.

   **Before (default: Simple executor active, Approval executor commented out)**

```
<ApplicationCreation
  executor="org.wso2.carbon.apimgt.impl.workflow.ApplicationCreationSimpleWorkflowExecutor"/>
<!--ApplicationCreation
  executor="org.wso2.carbon.apimgt.impl.workflow.ApplicationCreationApprovalWorkflowExecutor"/-->
```

4. Swap them: comment out the **Simple** executor and enable the **Approval** executor. Replace the two lines with:

   **After (copy this block)**

```
<!--ApplicationCreation
  executor="org.wso2.carbon.apimgt.impl.workflow.ApplicationCreationSimpleWorkflowExecutor"/-->
<ApplicationCreation
  executor="org.wso2.carbon.apimgt.impl.workflow.ApplicationCreationApprovalWorkflowExecutor"/>
```

   What changed: `<!--` and `-->` moved from the Approval line to the Simple line. Leave the rest of the file unchanged.

5. Click **Save Content**. The change takes effect immediately; no restart is needed.

## Part 2 – Create an Application

6. Log in to the Developer Portal, go to **Applications** → **Add New Application** and create an application:

   **Developer Portal URL**

```
https://<APIM_HOST>:<HTTPS_PORT>/devportal
```

| Field | Value |
|---|---|
| Application Name | `WorkflowTestApp` |
| Shared Quota | Keep the default |
| Description | Workflow test |

7. Click **Save**. Open **Applications** again: the status of `WorkflowTestApp` is **INACTIVE** (waiting for approval). You cannot generate keys or subscribe with it yet.

## Part 3 – Approve the Application

8. Log in to the Admin Portal and go to **Tasks** → **Application Creation**.

   **Admin Portal URL**

```
https://<APIM_HOST>:<HTTPS_PORT>/admin
```

9. Find the request for `WorkflowTestApp` and click **Approve** (or **Reject** to see the other outcome).

10. Back in the Developer Portal, refresh **Applications**:

| Action in Admin Portal | Application status |
|---|---|
| Approve | ACTIVE – keys can be generated and APIs subscribed |
| Reject | REJECTED – the application cannot be used |

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## Part 4 – Clean Up

**11.** Return to automatic approval: open the same registry file again (steps 1–2), put the two lines back as shown in the **Before** block of step 3, and click **Save Content**.

**12.** Optionally delete `WorkflowTestApp` in the Developer Portal.

## Expected Outcome

Application creation is controlled by an approval workflow: new applications stay INACTIVE until an administrator approves them in the Admin Portal, and become ACTIVE (or REJECTED) after the decision.

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